size_balanced_legacy

equity strategy | family: equity-baseline | status: reference | derived from `size_balanced`

What this strategy does

Extreme Picks, with an equal budget for each of the five company sizes -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. PEAD is stronger in small caps; balancing by size keeps the edge but removes a size bet.

Design notes (from the strategy's source file).

Size Balanced -- Extreme Picks with an equal budget for each company-size group.

Extreme Picks' trade, but each leg's budget is spread evenly over the size quintiles present that quarter, so micro-caps (where most extreme surprises happen) cannot dominate the book.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

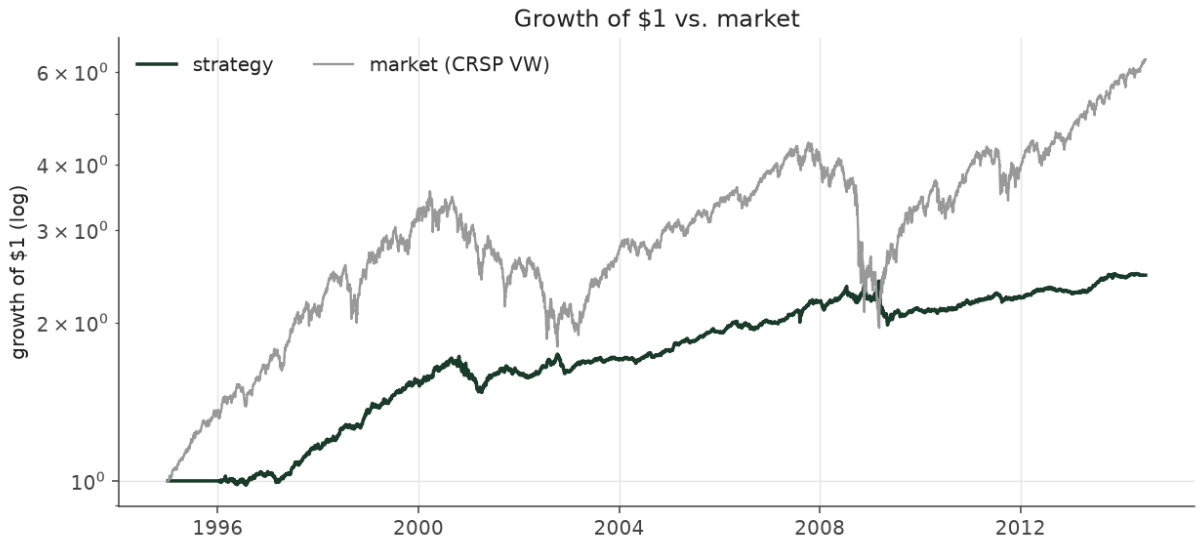
metric	size balanced legacy	market (CRSP VW)
Annualized return	4.74%	9.94%
Annualized volatility	5.26%	19.35%
Sharpe (NAV-pct)	0.91	0.59
Sortino	1.26	
Max drawdown	-17.5%	-55.5%
Calmar	0.27	
Final NAV ($10M start)	$24,622,256	
Worst day	-4.30%	
% positive days	52.2%	
Sharpe 1996-2001	1.32	
Sharpe 2002-2007	1.28	
Sharpe 2008-2013	0.34	

Correlation of daily returns with the market: **0.00**.

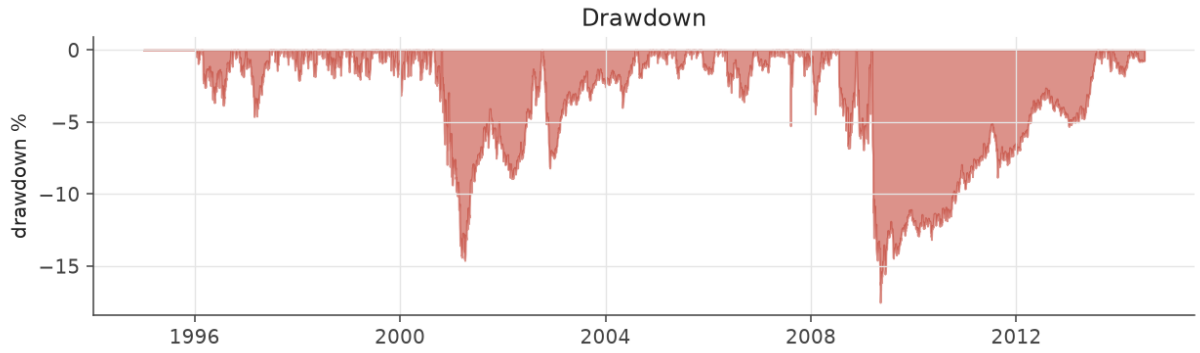
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 0.906

• best return among equity: `market_boost_max` (0.116); this one: 0.0474

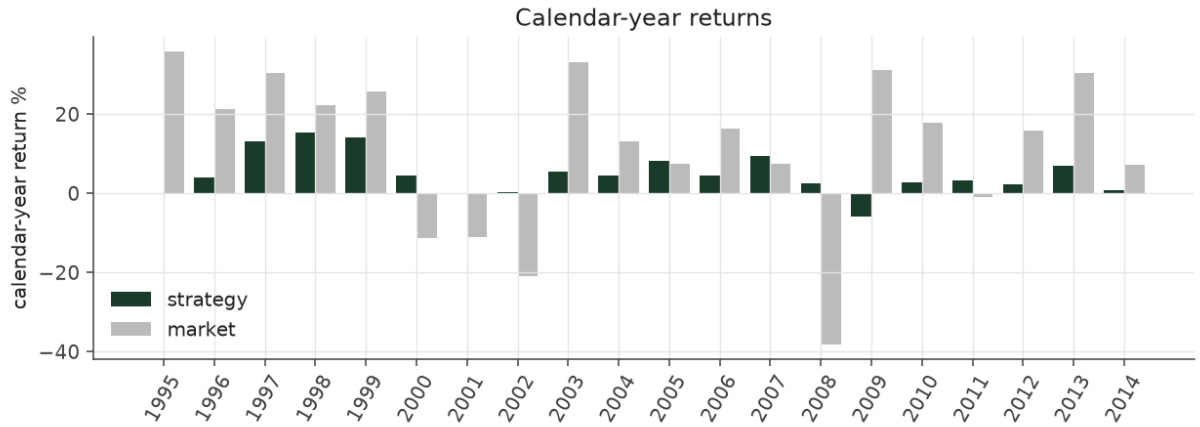
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.27



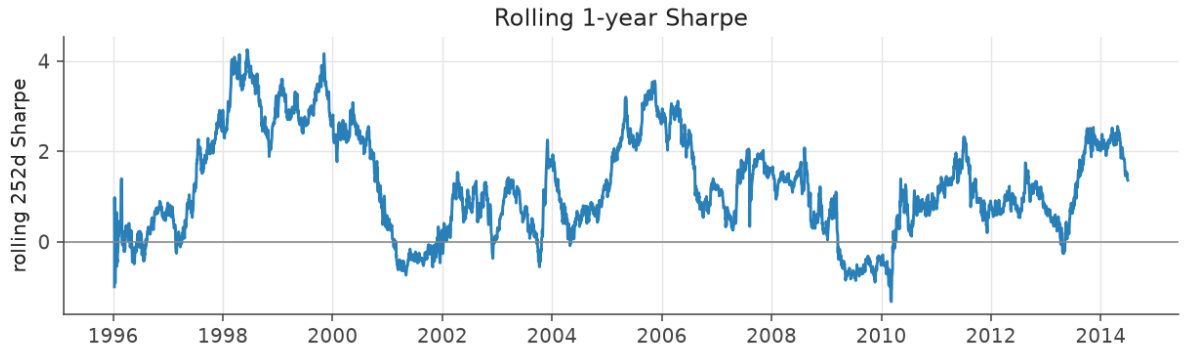
NAV (log scale)



Drawdown from running peak

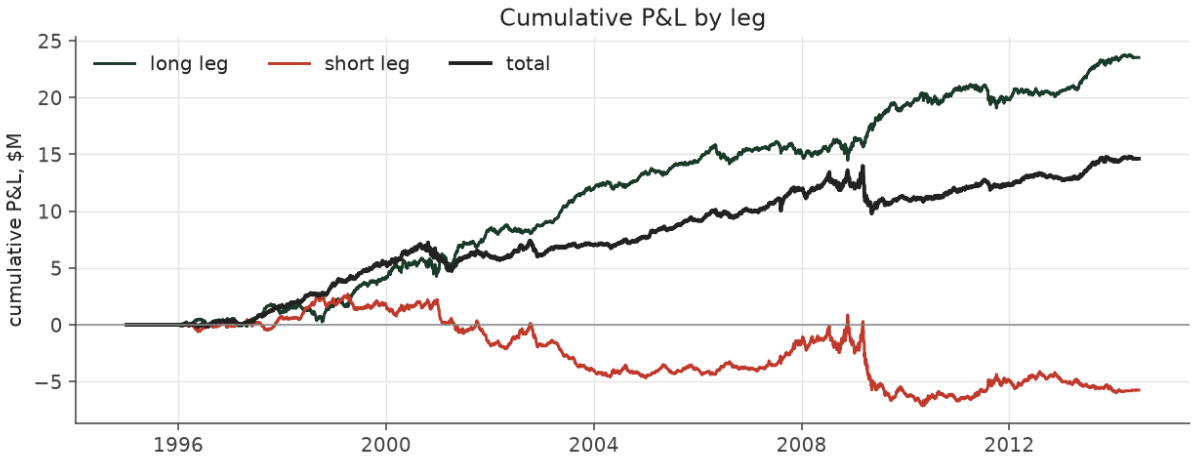


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	23.53	-5.75
P&L after costs ($M)	22.05	-7.43
of which common (average-event) component ($M)	11.79	-12.31
of which excess / PEAD component ($M)	11.52	6.84
excess minus costs ($M)	10.05	5.17
gross return per $ traded (bp)	309.93	-70.95
positions in the money (hit rate)	0.57	0.49
stand-alone Sharpe (leg only)	1.16	-0.23

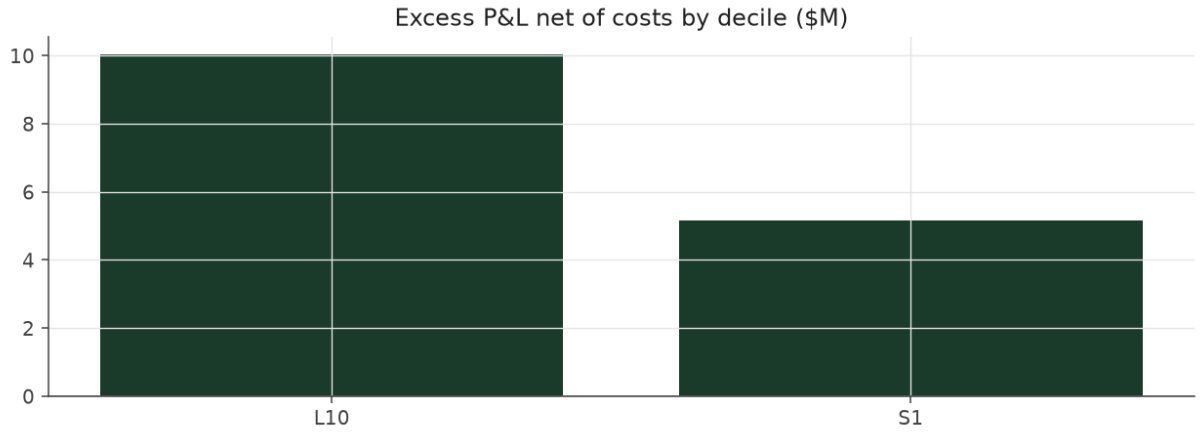
Daily P&L correlation between the two legs: **-0.64**.

• **The short leg loses money in absolute terms** (-5.7M before costs) while the long leg earns +23.5M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 11.8M / 12.3M. Stripping that common component, the surprise signal itself earned +11.5M on the long side and +6.8M on the short side.

• The two legs' daily P&L are -0.64 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.16 long / -0.23 short vs 0.91 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	10	17,174	759.10	22.05	11.52	1.48	10.04	66.0%	132.28
short	1	17,169	809.89	-7.43	6.84	1.68	5.16	33.9%	63.71



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	4,832	149.12	2.16	3.46	0.54	2.92	19.2%	195.53
2	7,159	327.44	4.51	5.52	0.90	4.62	30.4%	141.06
3	7,735	358.82	5.24	5.68	0.74	4.95	32.5%	137.85
4	7,731	365.93	1.70	2.24	0.57	1.67	11.0%	45.67
5	6,886	367.69	1.01	1.46	0.41	1.05	6.9%	28.52

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	4,357	93.37	2.65	3.32	0.47	2.85	18.8%	305.74
2	7,223	337.61	3.47	5.08	1.01	4.07	26.7%	120.39
3	7,834	386.64	5.77	6.92	0.77	6.15	40.4%	159.06
4	7,850	385.18	1.56	1.96	0.54	1.42	9.3%	36.90
5	7,077	366.08	1.20	1.10	0.37	0.73	4.8%	20.03

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	6,338	335.10	-0.26	3.22	0.51	2.71	17.8%	80.89
41-60d	11,202	486.79	1.49	2.28	0.99	1.29	8.5%	26.50

61-90d	12,665	583.62	5.83	5.16	1.17	3.99	26.2%	68.30
>90d	4,138	163.48	7.56	7.70	0.48	7.21	47.4%	441.30

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	9,920	371.88	5.68	6.59	0.71	5.88	38.6%	158.13
2002-07	11,506	527.09	5.64	6.76	1.04	5.71	37.6%	108.41
2008-13	12,917	670.02	3.30	5.01	1.41	3.61	23.7%	53.82

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	6,718	298.31	2.57	1.99	0.59	1.40	9.2%	46.90
Chems	687	31.76	-0.19	-0.12	0.06	-0.18	-1.2%	-57.47
Durbl	869	38.28	0.26	0.50	0.08	0.42	2.7%	109.15
Enrgy	1,375	66.03	1.65	1.81	0.13	1.67	11.0%	253.51
Hlth	3,037	133.90	1.35	1.10	0.29	0.81	5.3%	60.51
Manuf	3,627	163.12	0.76	1.40	0.32	1.08	7.1%	66.19
Money	5,366	255.16	1.65	3.65	0.50	3.15	20.7%	123.40
NoDur	1,540	71.33	0.94	1.00	0.14	0.86	5.7%	121.30
Other	5,074	228.66	2.19	2.66	0.51	2.15	14.2%	94.25
Shops	3,547	160.84	2.53	2.99	0.32	2.67	17.6%	166.20
Telcm	1,164	55.57	0.74	1.01	0.11	0.90	5.9%	162.04
Utils	1,339	66.02	0.17	0.37	0.11	0.26	1.7%	39.41

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	11,517	531.47	2.60	4.04	0.96	3.08	20.2%	57.96
low vol	11,554	516.11	6.39	7.82	1.15	6.67	43.9%	129.31
mid vol	10,730	492.43	5.30	6.13	0.99	5.14	33.8%	104.38

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	4.74%	0.00%
0.5%	0.9	4.53%	0.20%
1.0%	1.8	4.32%	0.41%
2.0%	3.6	3.88%	0.85%
5.0%	9.1	2.29%	2.44%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.